

CHRIS SCHMIDT

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PROFESSIONAL SUMMARY

Machine Learning Engineer dedicated to modernizing federal technology and advancing AI implementation across defense and government agencies. Currently supporting critical DoD data modernization initiatives with an active security clearance. Master's degree in Applied Mathematics; pursuing an MS in AI Engineering at Johns Hopkins University. Combines production ML/data engineering experience (Python, AWS GovCloud, PyTorch/TensorFlow) with self-directed depth in modern AI engineering practice, including verified agentic systems, computer vision, and RAG/evals, built through a portfolio of original projects. Passionate about applying LLMs, generative AI, and modern data architectures to complex civic and defense challenges at scale.

TECHNICAL SKILLS

Programming Languages: Python, SQL, R, SAS (professional); C++, Rust, Go, TypeScript/JavaScript, Shell scripting, Astro (personal projects)

Cloud Platforms: AWS GovCloud (Glue, S3, Athena, Lambda, RDS, DynamoDB, Kinesis, EFS, SQS, Elasticsearch), Azure, GCP, Databricks, Advana

Data & AI Engineering: ETL/ELT Pipelines, Data Ingestion & EDA, PowerBI, Qlik Sense, RAG architectures, RAGAS evaluation, vector search (pgvector/Supabase), evals-driven development, MLOps

Machine Learning: PyTorch, TensorFlow, Scikit-learn, BERT, NLP, OCR, Deep Learning, computer vision (YOLO, ByteTrack, TensorRT), Kalman filtering, DBSCAN, IsolationForest, geometric deep learning (exploratory)

AI/GenAI & Agentic Tooling: LLM application development, LangChain, LangGraph, Claude API, Claude Code, prompt engineering, agentic harness & multi-agent system design, Cursor, Codex

DevOps & Infrastructure: Docker, Kubernetes, Git/GitHub Actions, GitLab CI/CD, Terraform (IaC, exploratory), Fly.io, Vercel, Railway

Frameworks & Collaboration: FastAPI, Next.js, React/React Native (Expo), Supabase, Linux/Bash, Jira, Confluence, SharePoint

KEY GOVERNMENT PROJECTS

F-35 Data Modernization (DoD/JPO): Leading analytics work to achieve audit-worthy data architectures for the Joint Program Office, developing Python-based analysis tools and executive dashboards.

FORTIS AI Program (DoD): Developed data processing pipelines and integrated BERT-based NLP models to enhance AI capabilities for defense applications.

Advana Platform Analytics (DoD): Conducted enterprise-scale data quality analysis using Databricks and Qlik Sense, collaborating with MILDEP and IA stakeholders.

Center for Program Integrity (CMS): Performed big data analytics for fraud detection and program integrity using SAS and Python.

PERSONAL AI ENGINEERING PROJECTS

Self-directed systems built on one idea: keep consequential decisions in deterministic code, and independently check anything a model produces before it is trusted. Live demos, evaluations, and write-ups at pcschmidt.github.io.

SiteWatch AI: Computer Vision Safety Monitoring, Verified by an RLM Agent

- Built a two-plane safety monitor: YOLO11s + ByteTrack rules fire on every frame; a Prime Agent RLM verifies each firing by running Python over the raw tracks, and a deterministic gate recomputes its numbers before anything is stored
- Agent eval on 30 labeled incidents: 86.7% first pass at ~\$0.002 per incident; the gate caught a real agent misclassification on the live demo; 191 tests; Terraform validated for AWS, GCP, and Azure

Chokepoint: Provenance-Aware Transportation Intelligence

- Deterministic maritime, air, and land metrics over reviewed geofences, with an agent that answers only from evaluator-approved evidence: 25/25 on a labeled groundedness suite, 341 offline tests, live demo

Meridian: Agent Harness for AI-Assisted Development

- Harness that blocks an AI coding agent's next action when a gate fails, with an independent evaluator subagent to prevent self-grading (self-score 5.5/10 vs independent 2.5/10); 240 assertions across 18 test suites

Hard Power Intelligence: Defense/AI/Energy Intelligence Platform

- Multi-desk platform (FastAPI, Next.js, Supabase/pgvector) turning USAspending, SEC EDGAR, and NRC data into source-attributed briefings behind a fabrication-checking eval gate
- Entity resolver at 1.0 precision / 1.0 recall; passed a full OWASP Top 10 audit with zero Critical/High findings

CERTIFICATIONS

- AWS Solutions Architect Associate

- DeepLearning.AI Data Engineering Professional Certificate
- IBM Generative AI Engineering Certificate
- IBM AI Engineering Professional Certificate
- Deep Learning Specialization Certificate
- ML Ops | Machine Learning Operations Specialization, Duke University

SECURITY CLEARANCE

Active Security Clearance (Details available upon request)

PROFESSIONAL EXPERIENCE

MISSION TECHNOLOGIES, A DIVISION OF HII, Arlington, Virginia (Remote)

Machine Learning Engineer

October 2024 – Present

- Lead analytics for F-35 data modernization initiative (JPO), developing Python-based tools and PowerBI dashboards
- Perform enterprise-scale data architecture analysis to facilitate DoD audit-readiness transformations
- Develop Python scripts for data analysis, generating actionable recommendations for stakeholders
- Leverage Amazon GovCloud stack for secure, compliant data processing and storage

Software Engineer

May 2022 – October 2024

- Led data analysis and quality initiatives for DSCA/DoD modernization, collaborating with MILDEP and IA SMEs to align strategic goals
- Conducted exploratory data analysis (EDA) and built visualizations using Databricks and Qlik Sense within the Advana platform
- Developed data pipelines for the FORTIS AI program using Python and SQL, including API development, enhancing operational efficiency
- Created Python tools for complex EDA incorporating OCR, language translation, geospatial analysis, and graph-based approaches (Cypher, SPARQL, RDF)

Data Engineer, Associate

July 2021 – May 2022

- Developed serverless ETL microservices using AWS Lambda, RDS, DynamoDB, Kinesis, S3, EFS, SQS, and Elasticsearch, processing and ingesting up to 1 million text documents per hour
- Integrated BERT-based NLP models for the FORTIS AI project, enhancing language processing capabilities
- Conducted cost optimization analysis, reducing operational expenses
- Authored technical documentation and maintained security compliance requirements

PRIOR PROFESSIONAL BACKGROUND

Business Owner, Mobtown Restoration Group

8 years

- Owned and operated a construction/restoration firm, growing annual revenue to \$1–2 million and leading a team of up to 10 employees at peak

Senior Sales Associate, Institutional Fixed Income Sales, Wachovia Securities

6 years | Series 7/63

- Managed institutional client relationships and executed fixed income sales within a regulated financial services environment

Technical Writing Consultant, Medical Textbook & Journal Publishers

4 years

- Provided technical and medical/scientific writing services on a consulting basis, supporting publisher overflow workload

Mortgage Loan Originator

2 years

- Originated mortgage loans, performing credit analysis and underwriting support for borrowers

EDUCATION

Johns Hopkins Whiting School of Engineering

Master of Science, AI Engineering (Part-time, Expected May 2029)

Towson University

Master of Science, Applied Mathematics (Concentration: Applied Statistics)

Bachelor of Science, Applied Mathematics

Bachelor of Science, Political Science and History

LEADERSHIP

20 years as collegiate rugby head coach, including 10 years at Towson University and 10 years at Loyola University Maryland, focused on team development, strategy, and mentorship.